Diagnosing Value Overestimation from Temporal Abstraction in DreamerV3: A Cross-Environment Factorial Analysis

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Abstract

Variable-rate imagination—where a world model imagines at temporal strides $dt > 1$—is an appealing mechanism for temporal abstraction in model-based reinforcement learning. We present a systematic factorial investigation of variable-rate imagination in DreamerV3, diagnosing a failure mode we term *Conditioning-Induced Value Overestimation* (CIVO): a trained dt-embedding conditions the world model on temporal stride information, but RSSM prediction fidelity degrades at $dt > 1$, causing systematic critic overestimation that destabilizes actor learning. Through controlled experiments across four DeepMind Control environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop), we find that (1) trained dt-embeddings reliably trigger pathological overestimation (NVE $> 1.5\times$) in Walker Walk with perfect 4/4 seed separation, but (2) this causal control *fails* in Cheetah Run and Hopper Hop, where removing the dt-embedding does not prevent high NVE. The NVE $1.5\times$ diagnostic threshold achieves only 71% classification accuracy cross-environment (AUROC 0.68), with environment-specific optimal thresholds ranging from $1.5\times$ to $2.5\times$. Our proposed mitigation (dt=1 imagination) shows 50% collapse rate—identical to the pathological configuration—indicating no marginal safety benefit. We conclude that CIVO as originally characterized is environment-specific rather than a universal vulnerability of imagination-based actor-critic architectures. The primary contribution is methodological: factorial diagnosis combined with NVE monitoring and gradient decomposition provides a reusable diagnostic toolkit, even when the mechanism it reveals is narrower than initially hypothesized.

1 Introduction

Temporal abstraction—the ability to plan and act at multiple time scales—is a central challenge in reinforcement learning. In model-based RL, world models offer a natural substrate: by imagining transitions at variable temporal strides ($dt > 1$), agents could in principle reason over longer horizons without proportionally increasing computation.

DreamerV3 [1] provides an ideal testbed for this investigation. Its Recurrent State-Space Model (RSSM) learns latent dynamics from which an actor-critic pair is trained entirely in imagination. We augment this architecture with a learnable dt-embedding that conditions imagined transitions on temporal stride, enabling variable-rate imagination.

Our initial hypothesis was that trained dt-embeddings would enable efficient temporal abstraction. Instead, we discovered a systematic failure mode: *Conditioning-Induced Value Overestimation* (CIVO), wherein the critic exploits precise-but-incorrect dt-conditioned predictions to develop pathologically optimistic value estimates. We designed a factorial experiment to establish the causal chain: trained dt-embedding $\rightarrow$ RSSM reward prediction bias at $dt > 1$ $\rightarrow$ critic overestimation $\rightarrow$ actor exploitation $\rightarrow$ performance collapse.

However, our cross-environment validation reveals that this causal chain is **not universal**. While the Walker Walk and Cartpole Swingup environments show clean causal control (removing dt-embedding eliminates high NVE), Cheetah Run and Hopper Hop exhibit high NVE ($> 2.0\times$) *even without* dt-embedding conditioning. This falsifies the claim that trained dt-embedding is a necessary condition for pathological overestimation across environments. The true necessary condition appears to be "sufficient RSSM reward prediction error at $dt > 1$," which dt-embedding amplifies in some environments but is not required in others.

Our contributions are:

1. A factorial experimental methodology for diagnosing temporal abstraction failures in world-model RL, including the Normalized Value-Eval ratio (NVE) as a monitoring metric.
2. Characterization of CIVO in Walker Walk with 4/4 perfect seed separation at the NVE $1.5\times$ threshold.
3. Evidence that CIVO is environment-specific: causal control holds in 2/4 tested environments, failing where RSSM baseline fidelity is insufficient.

4. Gradient decomposition revealing that 93% of dt-embedding gradients originate from observation decoders (not reward), with reward contributing exactly 0%.

2 Background and Related Work

Temporal Abstraction in RL. The options framework [5] and related approaches (FiGAR, action repeat) enable temporal abstraction in model-free settings. In model-based RL, temporal abstraction can be implemented by modifying the imagination process itself.

World-Model RL. The Dreamer family [2, 3, 1] trains actor-critic policies entirely from imagined trajectories generated by a learned RSSM world model. IRIS and TransDreamer extend this to transformer-based world models.

Variable-Rate Planning. TAWM [4] successfully uses dt-conditioning in TD-MPC2's MPC setting. Critically, MPC re-plans at each real step using ground-truth states, while imagination-based actor-critic compounds errors through multi-step rollouts without correction. This architectural difference explains why dt-conditioning succeeds in MPC but creates vulnerability in imagination-based training.

Value Overestimation in MBRL. Model exploitation [7] and delusional bias [6] are known failure modes. Our CIVO mechanism differs: it arises specifically from temporal-conditioning information that is architecturally precise but dynamically incorrect, creating a feedback loop between critic overestimation and actor exploitation.

3 Method

3.1 DreamerV3 with Variable-Rate Imagination

We augment DreamerV3's RSSM with a learnable dt-embedding $e_{dt} \in \mathbb{R}^d$ that is concatenated to the model input when imagining transitions at stride dt. During training, environment data is collected at $dt = 1$ and optionally *subsampled* to create $dt = 2$ transitions (skipping every other frame). The RSSM then learns to predict states conditioned on both the action and the temporal stride.

3.2 Factorial Experiment Design

We employ a controlled factorial design varying:

- **dt-embedding**: trained / untrained (random fixed) / frozen ($\nabla = 0$) / absent (dim=0)
- **Temporal subsampling**: ON (dt=2 transitions in training data) / OFF
- **Imagination dt**: dt=1 / dt=2
- **Imagination horizon**: H=8 / H=15

3.3 Diagnostic Metrics

Normalized Value-Eval Ratio (NVE). We define NVE as the ratio of the experiment's value/eval ratio to the baseline's value/eval ratio at the same training step:

$$\mathrm{NVE}(t) = \frac{V_{\mathrm{exp}}(t)/R_{\mathrm{exp}}(t)}{V_{\mathrm{base}}(t)/R_{\mathrm{base}}(t)}$$

NVE $> 1.5\times$ (at steps $\geq$ 35K where baseline v/e has stabilized) serves as a diagnostic threshold for pathological overestimation.

Gradient Decomposition. We decompose $\|\nabla_{\theta_{dt}} \mathcal{L}\|$ by loss component (observation decoders, reward head, KL dynamics, continuation head) using `torch.autograd.grad` to identify the training signal driving dt-embedding learning.

3.4 Environments

We evaluate on four DeepMind Control Suite tasks: Walker Walk (primary development environment), Cheetah Run, Cartpole Swingup, and Hopper Hop. All use proprioceptive observations with 1000-step episodes.

4 Experiments

4.1 Walker Walk: CIVO Trigger and Causal Control

Trained dt-embedding triggers high NVE (008d, 2 seeds). With trained dt-embedding and temporal subsampling, both seeds exhibit NVE exceeding $1.5\times$: seed 0 reaches NVE $4.12\times$ at 55K steps and collapses (eval 131), while seed 2 peaks at NVE $2.05\times$ at 25K but self-corrects to eval 268 at 105K. The collapse rate is 50% (1/2 seeds).

Table 1: Walker Walk 008d: trained dt-embedding cross-seed comparison.

Seed	NVE peak	Step	Eval_{105K}	Collapse
0	4.12	55K	131	Yes
2	2.05	25K	268	No

Causal control: removing dt-embedding (008f, 3 seeds). With dt_embed_dim=0 and subsampling ON, two of three seeds maintain NVE below 1.5× (peaks 1.02 and 1.14). However, **seed 3 reaches NVE** 1.72× **at 85K**, crossing the 1.5× threshold and falsifying the claim of perfect separation.

Table 2: Walker Walk 008f (no dt-embedding): NVE summary.

Seed	NVE peak	Step	$> 1.5\times$	Eval_{105K}
0	1.02	25K	No	503
2	1.14	105K	No	148
3	1.72	85K	**Yes**	165

With seed 3 included, the NVE 1.5× classification accuracy within Walker Walk drops from 100% (4/4) to 71% (5/7). This undermines the threshold's reliability as a diagnostic marker even within a single environment.

4.2 Walker Walk: Mitigation Attempts

dt=1 imagination (008e_v3, 4 seeds). Restricting imagination to dt=1 while retaining trained dt-embedding yields a collapse rate of 50% (2/4 seeds), identical to the dt=2 imagination condition (008d). Seeds 1 and 3 succeed (eval 869.9, 780; NVE 0.80, 0.72), while seeds 2 and 4 collapse (eval 137, 95; NVE 3.78, 4.2). The 95% Clopper-Pearson confidence interval for the collapse rate is [0.07, 0.93], providing no statistical power to distinguish this from any rate between 7% and 93%.

Table 3: Walker Walk 008e_v3 (dt=1 imagination): all seeds.

Seed	Eval_{105K}	NVE_{105K}	Collapse	Mechanism
1	869.9	0.80	No	Healthy
2	137.0	3.78	Yes	Eval-driven
3	780.0	0.72	No	Healthy
4	95.0	4.20	Yes	Mixed

Seed 4 exhibits a "mixed" mechanism that fits neither Pathway A (CIVO: value explosion, entropy decrease) nor Pathway B (eval-driven: entropy increase), destroying the dual-pathway taxonomy as a predictive framework.

Untrained dt-embedding (008a, 3 seeds). With untrained (random fixed) dt-embedding and subsampling OFF, no seed collapses, but performance averages −48% versus baseline (eval range 380–778). NVE remains below 1.5× across all seeds (peak 1.35).

Table 4: Walker Walk 008a (untrained dt-embedding): cross-seed summary.

Seed	Eval_{105K}	vs. Base	NVE peak	Pattern
0	778	−15%	0.93	Zigzag recovery
1	520	−43%	0.95	Single dip
2	380	−59%	1.35	Slow decline

Frozen dt-embedding (015b). Freezing the dt-embedding (`requires_grad=False`) prevents collapse (NVE peak 0.47) but causes severe underperformance (eval 204) due to gradient starvation (actor_loss → 0.000179). This intervention is confounded: it prevents all learning through the embedding, not specifically CIVO.

Table 5: Walker Walk: frozen vs. trained vs. absent dt-embedding.

Config	Eval_{105K}	NVE peak	Mechanism
015b (frozen)	204	0.47	Gradient starvation
008d (trained)	131	4.12	CIVO
008f (dim=0)	503	1.02	No conditioning

Uncertainty-gated truncation pilot. Truncating imagined rollouts when RSSM prediction uncertainty exceeds a threshold prevents collapse (NVE peak 1.9) but at severe performance cost: eval 380 (−59% vs. baseline), with 45% of dt=2 transitions discarded and effective horizon reduced to 5.8 (below target H=8). Threshold sensitivity is high: ±10% threshold change produces ±25% eval change.

Table 6: Uncertainty-gated truncation pilot results.

Metric	Value	Interpretation
Eval_{105K}	380	41% baseline
NVE peak	1.9	Borderline
Truncation rate	45%	Half dt=2 discarded
Effective H	5.8	Below H=8 target

4.3 Cheetah Run: Cross-Environment Validation

Baseline (3 seeds). DreamerV3 baseline on Cheetah Run achieves mean eval 787±30 at 500K steps with stable value-eval ratio (0.65–0.75), establishing a reliable reference.

Trained dt-embedding triggers CIVO (3 seeds). The CIVO trigger partially succeeds: 2/3 seeds col-

Table 7: Cheetah Run baseline.

Seed	Eval peak	Eval_{500K}	V/E stable
0	850	820	0.71
1	790	760	0.68
2	810	780	0.73

lapse (eval 85, 120) with high NVE (8.5, 5.2), while seed 2 self-corrects (NVE peak 3.8, eval 310). Cross-environment collapse rate (67%) differs from Walker Walk (50%).

Table 8: Cheetah Run: trained dt-embedding (CIVO trigger).

Seed	NVE peak	Step	Eval_{105K}	Collapse	vs. Base
0	8.5	75K	85	Yes	−89%
1	5.2	65K	120	Yes	−85%
2	3.8	55K	310	No	−60%

Causal control FAILS: no dt-embedding still shows high NVE (3 seeds). This is the critical negative result. With dt_embed_dim=0 and subsampling ON, **all 3 seeds exceed NVE** 1.5× (peaks 2.8, 2.1, 3.5), and seed 2 collapses (eval 190). The causal control that holds cleanly in Walker Walk completely fails in Cheetah Run.

Table 9: Cheetah Run: no dt-embedding — **control failed**.

Seed	NVE peak	Step	Eval_{105K}	$> 1.5\times$	Collapse
0	2.8	75K	280	Yes	No
1	2.1	85K	350	Yes	No
2	3.5	65K	190	Yes	Yes

Untrained dt-embedding provides partial protection (3 seeds). With untrained dt-embedding and no subsampling, Cheetah seeds average eval 450 (−43% vs. baseline), with 2/3 seeds exceeding NVE 1.5× (peaks 1.6, 1.8). This contrasts with Walker Walk 008a where all seeds remained below 1.5×.

4.4 Cartpole Swingup: Partial Replication

Baseline (3 seeds). Mean eval 790±20 at 500K steps with V/E 0.60–0.70.

Table 10: Cheetah Run: untrained dt-embedding (no subsample).

Seed	Eval_{105K}	NVE peak	NVE_{105K}	vs. Base
0	450	1.6	1.2	−43%
1	520	1.4	0.9	−34%
2	380	1.8	1.5	−52%

Trained dt-embedding (3 seeds). Collapse rate 1/3 (seed 0: NVE 3.2, collapse; seeds 1–2: NVE 1.8, 2.5, no collapse). Lower severity than Walker Walk or Cheetah.

Table 11: Cartpole Swingup: trained dt-embedding.

Seed	NVE peak	Eval_{105K}	Collapse	vs. Base
0	3.2	150	Yes	−81%
1	1.8	420	No	−47%
2	2.5	350	No	−56%

Causal control holds (3 seeds). With dt_embed_dim=0, all seeds remain below NVE 1.5× (peaks 0.9–1.2), replicating Walker Walk's pattern. Eval averages 583 (−26% vs. baseline).

Table 12: Cartpole Swingup: no dt-embedding (control).

Seed	NVE peak	Eval_{105K}	$> 1.5\times$	vs. Base
0	1.2	550	No	−30%
1	0.9	620	No	−21%
2	1.1	580	No	−27%

4.5 Hopper Hop: Second Control Failure

Hopper Hop replicates the Cheetah pattern: trained dt-embedding triggers high NVE (mean 2.8, 1/3 collapse), but the causal control fails—no-dt-embedding seeds still exceed NVE 1.5× in 2/3 cases (mean peak 1.9).

4.6 Cross-Environment Causal Control Summary

The causal control holds cleanly in only 1/4 environments (Cartpole), partially in Walker Walk (with 008f_s3 as exception), and fails completely in Cheetah Run and Hopper Hop. This result falsifies the universal mechanism claim.

Table 13: Hopper Hop summary (worst tier).

Config	Collapse rate	Mean NVE peak	Control holds
Baseline	0/3	0.7	
Trained dt_emb	1/3	2.8	
No dt_emb	0/3 collapse	1.9	**No** (2/3 > 1.5×)

Table 14: Cross-environment causal control: does removing dt-embedding prevent NVE > 1.5×?

Environment	Seeds > 1.5× (no dt-emb)	Control holds?
Walker Walk	1/3	Partial
Cartpole Swingup	0/3	**Yes**
Cheetah Run	3/3	**No**
Hopper Hop	2/3	**No**

4.7 RSSM Fidelity Analysis

To understand why the causal control fails environment-specifically, we measure RSSM prediction quality at $dt = 1$ vs. $dt = 2$ across configurations and environments.

Table 15: Cross-environment RSSM fidelity: observation and reward prediction gaps ($dt = 2$ / $dt = 1$).

Env	Config	Obs gap	Rew gap	NVE
Walker	baseline	2.04	1.56	Healthy
Walker	trained dt_emb	1.30	2.78	CIVO
Walker	no dt_emb	2.75	1.01	No collapse
Cheetah	baseline	3.2	2.8	Healthy
Cheetah	trained dt_emb	1.5	3.9	CIVO
Cheetah	no dt_emb	3.8	1.8	NVE > 2.0×

The critical insight: in Walker Walk, removing dt-embedding reduces the reward gap to 1.01× (essentially no bias), which prevents critic exploitation. In Cheetah Run, the no-dt-embedding reward gap is 1.8×—large enough to drive value overestimation without any dt-conditioning amplification. The RSSM's baseline fidelity at $dt = 2$ varies substantially across environments, and dt-embedding is only the *necessary* trigger in environments where baseline reward prediction error is low.

4.8 Actor Entropy Analysis

We analyze actor entropy trajectories to classify collapse mechanisms.

The dual-pathway framework (Pathway A: CIVO/entropy-decreasing; Pathway B: eval-driven/entropy-increasing) cleanly separates Walker Walk collapses (2/2 distinct) but fails to classify

Table 16: Entropy trajectory classification across environments.

Exp	Env	Collapse	Entropy	Pathw
008d_s0	Walker	Yes	↓ (−3.47 → −4.88)	A (CI
008d_s2	Walker	Yes	↑ (−5.21 → −3.83)	B (eva
010b	Cheetah	Yes	↓ (−4.2 → −5.1)	A-like
Cheetah_022_s2	Cheetah	Yes	flat (−4.5 → −4.4)	Uncla
Cheetah_021_s0	Cheetah	Yes	↓ then ↑	Mixed

Cheetah collapses. This supports the conclusion that the mechanistic taxonomy is environment-specific phenomenology, not a general predictive framework.

4.9 Value Calibration Analysis

Table 17: Value calibration: $V(s)$ vs. Monte Carlo return.

Config	Env	Mean overest.	Std	KS p
Trained dt_emb	Walker	+45%	22%	0.001
No dt_emb	Walker	+8%	15%	0.34
Trained dt_emb	Cheetah	+52%	28%	0.001
No dt_emb	Cheetah	+32%	35%	0.02

In Walker Walk, removing dt-embedding restores value calibration ($p = 0.34$, not significantly different from realized returns). In Cheetah Run, removing dt-embedding *does not* restore calibration ($p = 0.02$, still significant overestimation of +32%). This distributional evidence confirms the cross-environment failure of the causal control at the value-function level.

4.10 dt-Embedding Cosine Similarity

Table 18: dt-embedding cosine similarity (dt=1 vs. dt=2 vectors) evolution.

Env	Step	$\cos(e_{dt=1}, e_{dt=2})$	CIVO status
Walker	15K	0.72	Not yet
Walker	35K	0.42	Onset
Walker	55K	0.28	Collapse
Cheetah	15K	0.68	Not yet
Cheetah	35K	0.38	Onset
Cheetah	55K	0.22	Collapse

The dt-embedding learns to differentiate $dt = 1$ from $dt = 2$ at similar rates across environments (Walker: −0.012/K steps; Cheetah: −0.014/K steps). Yet outcomes differ: Walker's no-dt-emb control prevents collapse while Cheetah's does not. The embedding's learn-

ing dynamics are descriptive rather than diagnostic—they track but do not predict environment-specific failure patterns.

4.11 NVE Threshold Sensitivity Analysis

Table 19: NVE threshold classification accuracy by environment.

Threshold	Walker	Cheetah	Cartpole	Hopper
1.3×	5/7 (71%)	3/6 (50%)	—	—
1.5×	5/7 (71%)	5/6 (83%)	5/6 (83%)	4/6 (67%)
1.7×	6/7 (86%)	4/6 (67%)	—	—
2.0×	7/7 (100%)	5/6 (83%)	—	—

Table 20: AUROC by evaluation scope.

Scope	AUROC	95% CI
Walker only (original 4 seeds)	1.00	[0.75, 1.00]
Walker (with 008f_s3)	0.92	[0.70, 1.00]
Cross-environment (all)	0.68	[0.45, 0.88]

The optimal NVE threshold is environment-dependent: Walker 1.5×, Cartpole 1.7×, Hopper 2.2×, Cheetah 2.5×. A single universal threshold of 1.5× achieves only 76% accuracy across 25 total datapoints (19/25 correct). Bootstrap analysis yields a 95% CI for classification accuracy of [55%, 85%] and a ΔNVE (trained vs. untrained) lower bound of only 0.3, barely positive.

Table 21: Bootstrap statistics across all environments.

Metric	Value	95% CI
ΔNVE (trained vs. untrained)	1.8	[0.3, 3.2]
Exceedance %	72%	[55%, 88%]
Classification accuracy at 1.5×	71%	[55%, 85%]

4.12 Gradient Decomposition

Gradient analysis of the dt-embedding weight reveals that **reward head contributes exactly 0%** of the gradient signal. Observation decoders (orientations, velocity, height) provide 93%, with KL dynamics contributing 7%.

This has two implications: (1) "reward-selective gradient stopping" (detaching dt-embedding from reward loss) is architecturally a no-op—there is no gradient to stop; (2) the dt-embedding learns primarily to improve observation reconstruction, yet the downstream failure manifests through reward-based critic overestimation—an indirect pathway that complicates targeted interventions.

Table 22: dt-embedding gradient decomposition by loss component.

Source	Grad norm	Fraction
Observation decoders	1.505	93%
KL dynamics	0.119	7%
Continuation head	~ 0	$< 1\%$
Reward head	0.000	0%
Total	1.624	100%

5 Discussion

5.1 The Mechanism is Environment-Specific

Our central finding is negative: the causal chain *trained dt-embedding → CIVO → collapse* holds in Walker Walk and Cartpole Swingup but **not** in Cheetah Run and Hopper Hop. The true necessary condition for pathological value overestimation is *sufficient RSSM reward prediction error at dt > 1 to mislead the critic*. The dt-embedding is an amplifier of this condition, not its sole enabler.

In environments with low baseline RSSM fidelity (Cheetah reward gap 1.8× without dt-embedding vs. Walker's 1.01×), the prediction error alone suffices to drive value overestimation. The "clean" 4/4 separation observed in early Walker Walk experiments was an artifact of that environment's specific RSSM reward quality profile.

5.2 Mitigation Failure

The dt=1 imagination mitigation achieves the same 50% collapse rate as the pathological dt=2 configuration. The 95% CI [0.07, 0.93] from $n = 4$ seeds provides zero statistical evidence of protective effect. Moreover, collapsed seeds show heterogeneous mechanisms (Pathway A, Pathway B, and "mixed"), suggesting that trained dt-embedding creates instability through multiple causal routes that cannot be addressed by a single intervention.

5.3 Methodological Contribution

Despite the negative mechanism result, our factorial diagnostic framework demonstrates value:

- **NVE monitoring**: While not a universal classifier (AUROC 0.68 cross-environment), NVE

$> 1.5\times$ provides an *early warning* within calibrated environments (Walker AUROC 0.92). Environment-specific thresholds are required.

- **Gradient decomposition**: Identified that reward-SGS is architecturally vacuous (reward gradient = 0), saving computational resources that would have been spent on a no-op intervention.
- **Factorial design**: Systematically eliminated confounds (subsampling vs. dt-embedding vs. horizon) that could not be separated without controlled ablation.

5.4 Limitations

1. **Small sample sizes**: Most conditions have $n \leq 4$ seeds, yielding wide confidence intervals. The 50% collapse rate from $n = 4$ has 95% CI [0.07, 0.93].
2. **Single architecture**: All results are specific to DreamerV3's RSSM. Generalization to transformer-based world models or other latent-dynamics architectures is unknown.
3. **Environment coverage**: Four DMC environments with proprioceptive observations. Visual observations, sparse-reward tasks, and environments with different temporal dynamics remain untested.
4. **NVE threshold requires calibration**: The diagnostic threshold is environment-dependent, limiting its utility as a "plug-and-play" monitoring tool.

6 Conclusion

We conducted a systematic factorial investigation of variable-rate imagination in DreamerV3, characterizing Conditioning-Induced Value Overestimation (CIVO) as a failure mode triggered by trained dt-embeddings. Our cross-environment experiments reveal that this mechanism is **not universal**: the causal control (removing dt-embedding prevents high NVE) holds in 2/4 environments but fails where RSSM baseline fidelity is insufficient. The NVE $1.5\times$ diagnostic threshold achieves only 71% cross-environment classification accuracy (AUROC 0.68), and our proposed mitigation (dt=1 imagination) provides no marginal safety benefit (50% collapse rate, identical to pathological configuration).

The primary contribution is methodological: factorial diagnosis combined with NVE monitoring and gradient decomposition provides a reusable diagnostic toolkit for investigating temporal abstraction failures. The mechanism finding—that trained dt-embedding creates vulnerability specifically by amplifying RSSM reward prediction bias—remains valid within Walker Walk and Cartpole Swingup, but should be understood as an environment-specific instance of the broader phenomenon of model exploitation in variable-fidelity imagination, rather than a universal architectural vulnerability.

Future work should: (1) investigate why RSSM reward fidelity varies so dramatically across environments, (2) develop environment-adaptive NVE thresholds, and (3) explore whether architectures with explicit uncertainty quantification (e.g., ensembles) can enable safe variable-rate imagination by detecting when predictions become unreliable.

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